

Android Studio Setup and Installation Guide (Windows 11)

1. System Requirements

Ensure your system meets the minimum requirements before installation:

- OS: Windows 10 or 11 (64-bit)
- RAM: 8 GB minimum (16 GB recommended)
- Disk Space: 8 GB free (SSD preferred)
- CPU: Intel i5/i7/i9 or AMD Ryzen
- Graphics: OpenGL 3.0+ support

2. Download Android Studio

1. Go to <https://developer.android.com/studio>
2. Click "Download Android Studio" and accept the terms.
3. Save the installer (e.g. android-studio-202x.x.x.exe).

3. Installation Steps

1. Run the installer.
2. Choose Standard Installation (recommended).
3. Select Android SDK, SDK Platform-Tools, and AVD.
4. Accept defaults and complete installation.

4. Configure SDK

1. Open Android Studio → More → SDK Manager.
2. Install:
 - Android SDK Platform (latest)
 - Android SDK Tools
 - Android Emulator
 - Intel x86 Emulator Accelerator (HAXM)
3. SDK path (default):
C:\Users\\AppData\Local\Android\Sdk

5. Configure Environment Variables

1. Open Environment Variables → System Variables → Path → Edit.
2. Add:
C:\Users\\AppData\Local\Android\Sdk\platform-tools
C:\Users\\AppData\Local\Android\Sdk\tools
C:\Users\\AppData\Local\Android\Sdk\tools\bin
3. Verify using CMD: adb --version

6. Setup Android Virtual Device (AVD)

1. Android Studio → Tools → Device Manager → Create Device.
2. Choose Pixel device and Android 14 (API 34) image.
3. Finish setup and start emulator.

7. Create First Project

1. New Project → Empty Activity.
2. Language: Java or Kotlin.
3. Minimum SDK: API 24+.
4. Click Finish → Wait for Gradle build.
5. Run  to launch app on emulator.

8. Optional Configurations

- Enable HAXM or Hyper-V acceleration (for faster emulator)
- Update SDKs and tools regularly
- Check Gradle JDK path under Settings → Build Tools → Gradle.

9. Troubleshooting

- Gradle sync failed → Check internet/proxy.
- Emulator slow → Enable virtualization in BIOS.
- ADB not found → Verify SDK paths.
- HAXM error → Run intelhaxm-setup.exe manually.

10. Verify Installation

Run these commands in CMD:

```
java -version
```

```
adb --version
```

```
emulator -list-avds
```

If all show outputs, setup is successful.